

POLLUTION

DEFINITION OF POLLUTION

- When Harmful Substances Contaminate the Environment it is Called Pollution.
- Pollution refers to the very bad condition of environment in terms of quantity and quality .

Types of Pollution

There are Five types of Pollution

- Air Pollution
- Water Pollution
- Noise Pollution
- Land Pollution
- Radio Active Pollution



What is Atmosphere

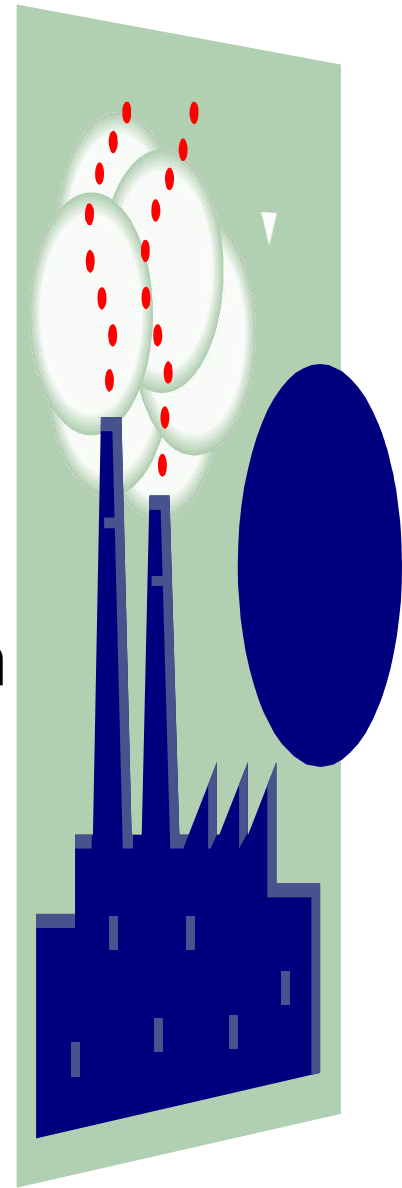
- Atmosphere is the life blanket of Earth.
- It is therefore essential that we know more about the atmosphere and the ways in which it is Polluted.



Causes of Air Pollution

Major sources of Air Pollution

- Industries
- Automobiles and Domestic fuels
- High Proportion of undesirable gases, such as sulphur dioxide and carbon monoxide



Effects of Air Pollution

Air Pollution affects???

- Human health
- Animals
- Plants
- The atmosphere as a whole



How to Avoid Air Pollution

Yes, we can avoid pollution As Follows



**Use natural Gases, like lpg autos
Do not Burst Crackers
Use less Amount of Fuel for Vehicles
Avoid using and use electric stoves
(bio gas)**

Water God's Valuable Gift



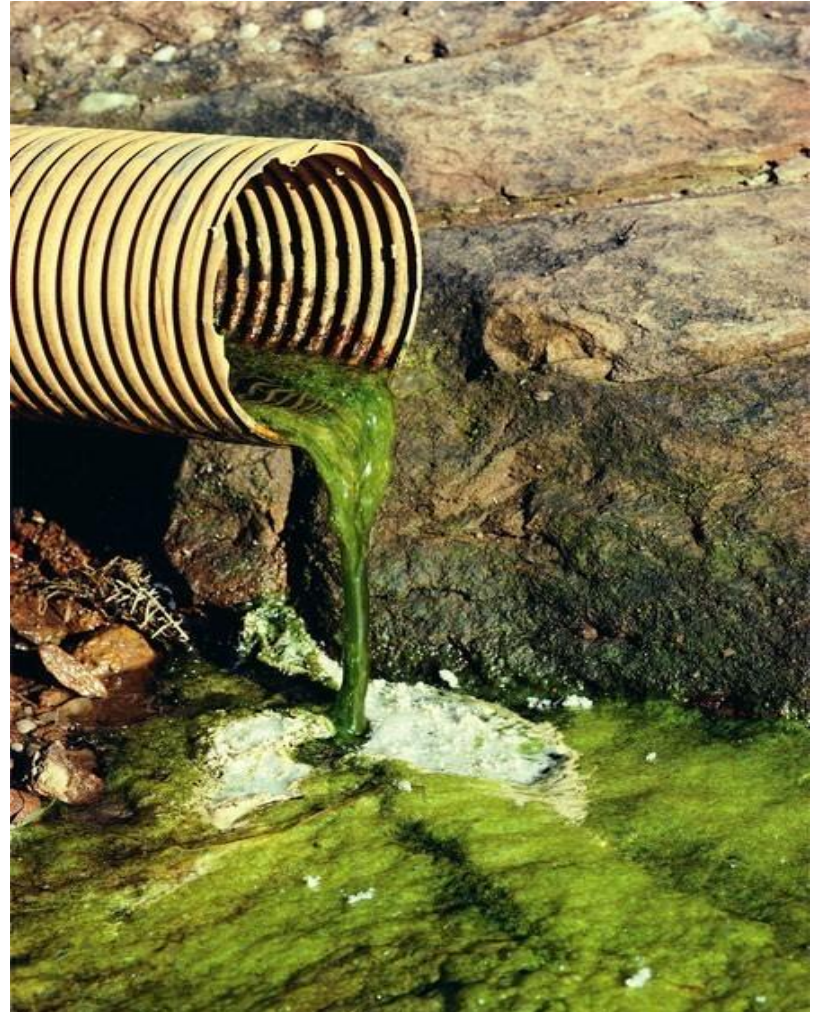
Definition of Water Pollution

The Contamination of water with undesirable substances which make it unfit for usage is termed water Pollution.

Causes of Water Pollution

- About 40% of Deaths worldwide are caused by Water Pollution.
- Water Pollution is Caused by organic and inorganic industrial wastes and effluents discharged into rivers.
- Noise also causes anxiety stress recreation and fright.

Water Pollution Pictures



Effects of Water Pollution

- Diseases like Cholera
- Malaria
- Typhoid (spread during the rainy season)
- Aquatic life gets destroyed

How to Avoid Water Pollution

- Rivers should not be used for washing clothes or bathing animals in.
- Harvesting of Rainwater to meet water requirements.
- Dams & embankments must be created.
- The rivers must not be contaminated.
- In sacred river like Ganga the dead bodies shouldn't thrown.

Definition of Land Pollution

- One fourth of area is covered by land is Called Land.
- Land is a earth which is occupied by people for shelter, occupation etc..,

Causes of Land Pollution



We can classify major sources that lead to land following Categories

- Mining and quarrying
- Sewage waste
- Household Garbage
- Industrial Waste

Land pollution pictures



Plastic



Chemical



Trash



Fertilization

Effects of Land Pollution

- The Land Cannot be construct house
- Man cannot be farming
- Ground water will gets Affected
- House hold Garbage like putting Plastics

Prevention of LandPollution

- More and more land should be brought under farming
- Trees should be planted everywhere.
- Waste matter should be disposed immediately
- Avoid drilling the Land for more underground water.
- Avoid using more fertilizers and Pesticides.

Dam

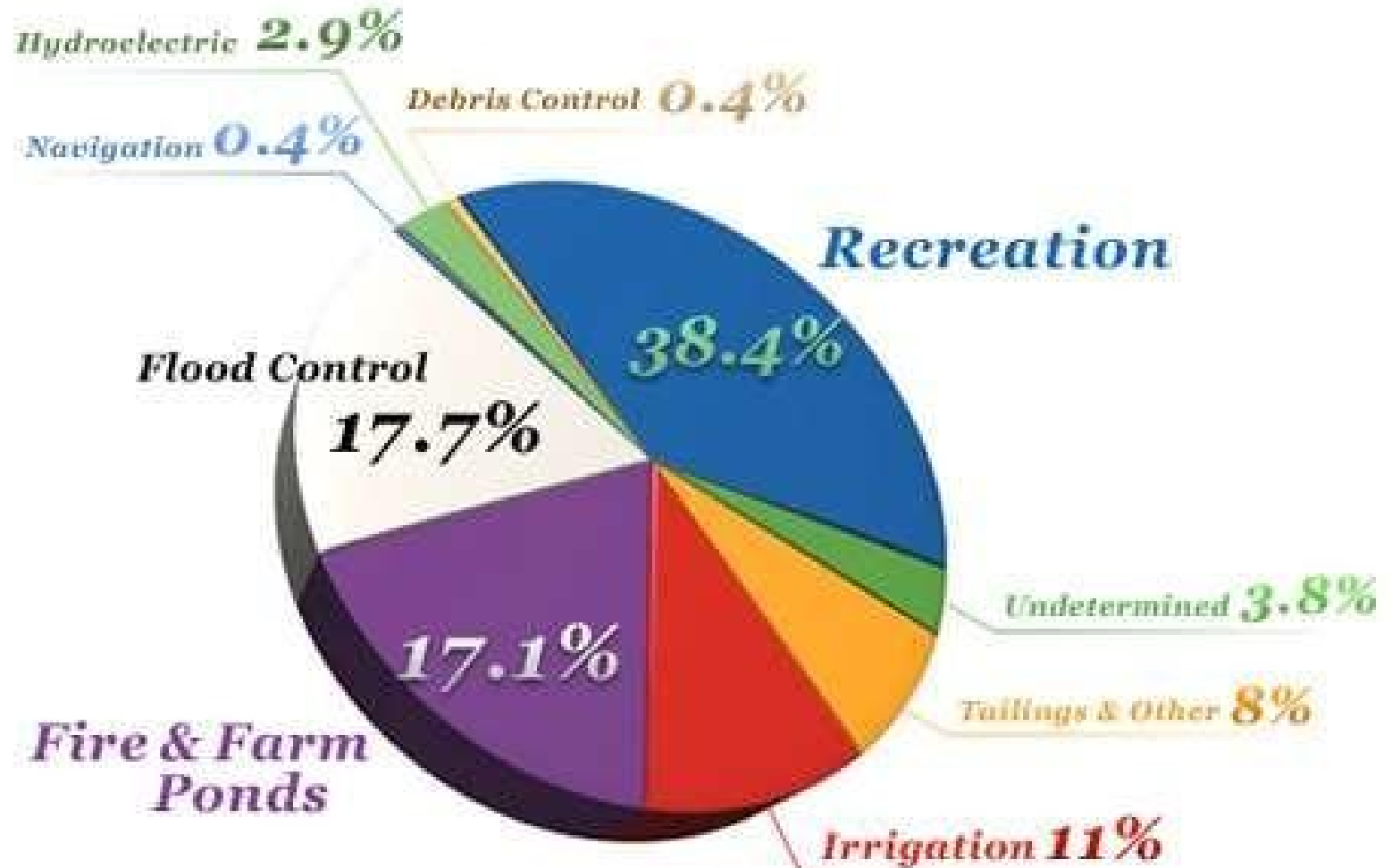
Dam is a solid barrier constructed at a suitable location across a river valley to store flowing water.



Storage of water is utilized for following objectives:

- ❑ Hydropower
- ❑ Irrigation
- ❑ Water for domestic consumption
- ❑ For drought and flood control
- ❑ Other additional utilization is to develop fisheries.

ADVANTAGES OF DAMS



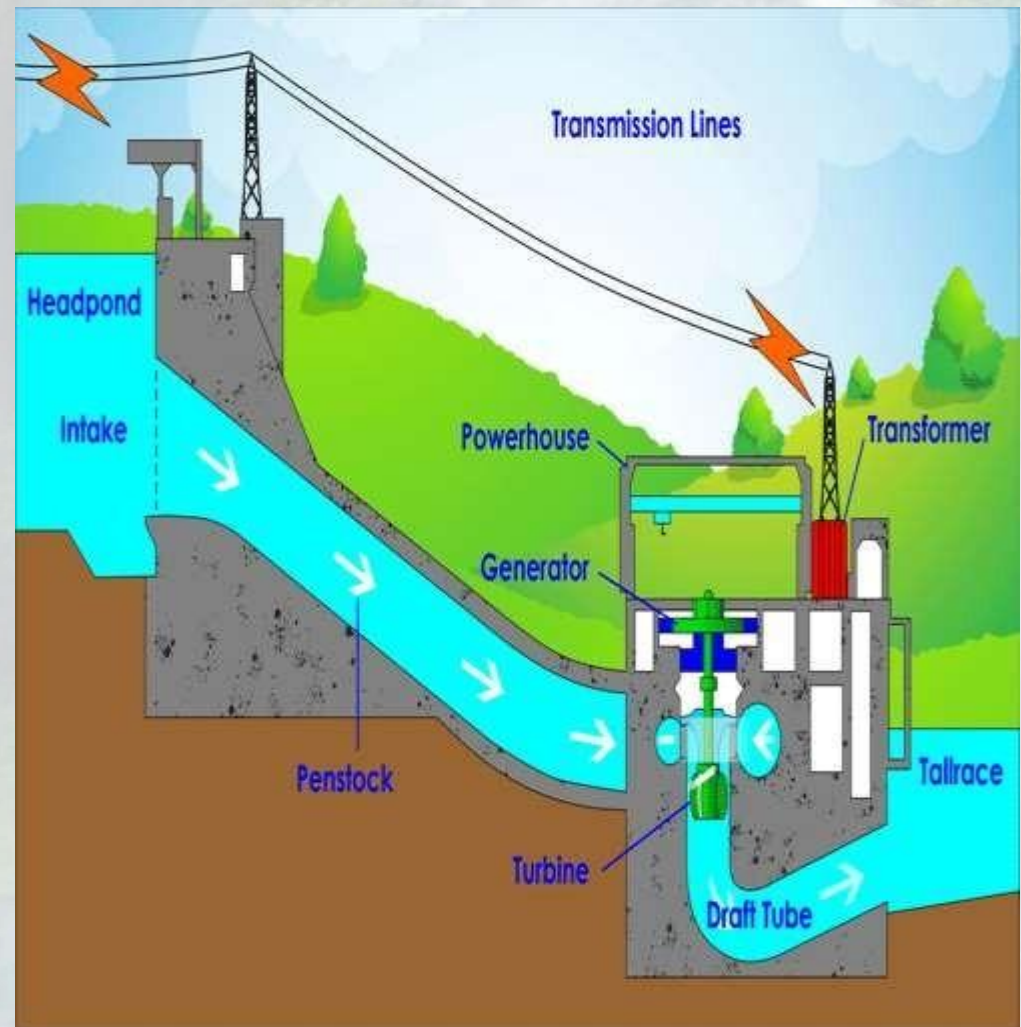
Drought Control

Dams are needed in places of permanent Drought. To act as barriers to hold drinking Water.



Hydropower generation

The most prominent use of large dams is for hydroelectric power generation, combined with a hydropower plant. Compared to other renewable sources, hydropower produces the most cost-efficient energy at costs as low as 2 US¢/kWh.



Irrigation for agriculture

Water stored in a reservoir is used to irrigate agricultural crops. It is estimated that 30-40% of irrigated lands worldwide, which contribute 12-16% of global food production, rely on dams.



Water supply

- Especially in urban areas, reservoir storage capacities are used to supply drinking water or water for industrial purposes.

Flood control

As a positive side effect, dam structures often offer flood protection by regulating water discharge and balancing runoff differences.



INLAND NAVIGATION

▲ The navigability of a water system enabled through the regulation by the dam may provide an impetus for trade and development. Inland navigation is the most cost effective and least polluting means of transportation.



Finally, the lakes created through water storage can additionally serve recreational and touristic purposes.

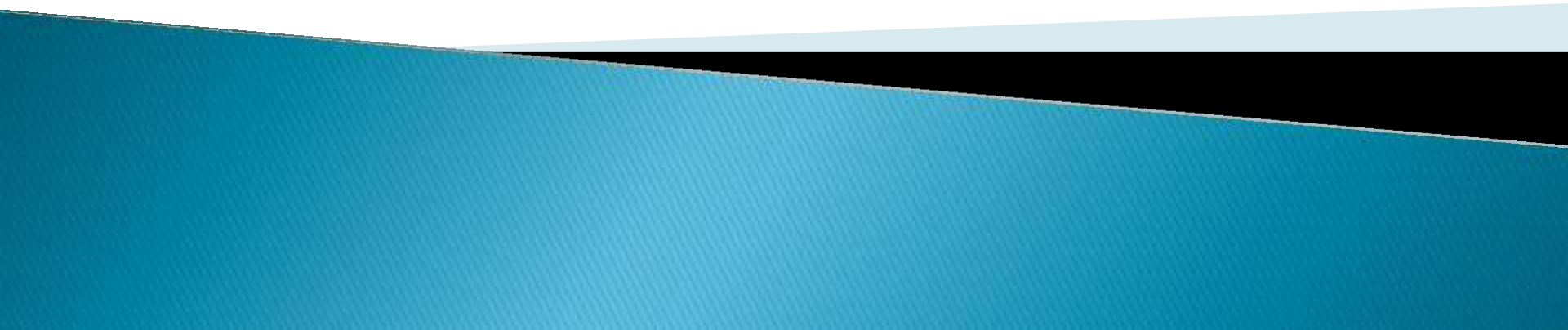


Flood and Flow Control

This is due to flood peaks being reduced, and the uniformity in base flow being increased throughout.

DISADVANTAGES

- Dams have endangered some species of fish
- Reservoirs can foster diseases if not properly maintained
- Reservoir water can evaporate significantly, Some researchers believe that reservoirs can cause earthquakes



Dams detract from natural settings, ruin nature's work



Dams have inhibited the seasonal migration of fish



Risk of failure

▲ when a dam fails it can be catastrophic.

- As recently as recently as 2004 the Big Bay Dam in Mississippi broke destroying nearly fifty homes. It is a major concern in civil engineering to see that dams are safe from hazards such as landslides and earthquakes.





conclusion

Dams are very costly projects,so their construction in seismic areas needs careful study to ensure their safety.when earth quakes occurs,a dam is subjected to two forces are due to the dam and due to reservoir water.



Solid Waste

- **Solid Waste** in general can be defined as ‘As a material which has negligible value to the producer and there is no direct consumption of the generated waste’.
- It is generated due to various activities that can be residual and commercial, agricultural, etc.
- In a nutshell “ Solid waste can be regarded as refuse or waste from any kind of source”. But any refuse or waste can be economic resource to others.

Basic terms related to solid waste

1. **Ash** : the non-combustible solid by-products of incineration or other burning process.
2. **Bulky waste**: large wastes such as appliances, furniture, and trees and branches, that cannot be handled by normal MSW processing methods.
3. **Co-disposal**: the disposal of different types of waste in one area of a landfill or dump. For instance, sewage sludges may be disposed of with regular solid wastes.

4. **Biodegradable material** : any organic material that can be broken down by microorganisms into simpler, more stable compounds. Most organic wastes (e.g., food, paper) are biodegradable.
5. **Compost** : the material resulting from composting. Compost, also called humus, is a soil conditioner and in some instances is used as a fertilizer.
6. **Composting** : biological decomposition of solid organic materials by bacteria, fungi, and other organisms into a soil-like product.

7. **Disposal** : the final handling of solid waste, following collection, processing, or incineration. Disposal most often means placement of wastes in a dump or a landfill.
8. **Environmental impact assessment (EIA)** : an evaluation designed to identify and predict the impact of an action or a project on the environment and human health and well-being. Can include risk assessment as a component, along with economic and land use assessment.

9. Environmental risk assessment (EnRA) : an evaluation of the interactions of agents, humans, and ecological resources. Comprised of human health risk assessment and ecological risk assessment, typically evaluating the probabilities and magnitudes of harm that could come from environmental contaminants.

10. Garbage : in everyday usage, refuse in general. Some MSWM manuals use garbage to mean "food wastes," although this usage is not common.

11. Landfilling : the final disposal of solid waste by placing it in a controlled fashion in a place intended to be permanent. The Source Book uses this term for both controlled dumps and sanitary landfills.

12. Leachate : liquid that has seeped through a landfill or a compost pile and has accumulated bacteria and other possibly harmful dissolved or suspended materials.

13. MSW : municipal solid waste.

14. MSWM : municipal solid waste management.

15. Putrescible : subject to decomposition or decay. Usually used in reference to food wastes and other organic wastes that decay quickly.

16. Refuse : all kinds of wastes in solid state excepting excreta from residential, commercial and industrial area.

17. Refuse-derived fuel (RDF) : fuel produced from MSW that has undergone processing. Processing can include separation of recyclables and non-combustible materials, shredding, size reduction, and pelletizing.

18. Rubbish : a general term for solid waste. Sometimes used to exclude food wastes and ashes.

19. Waste-to-energy (WTE) plant : a facility that uses solid waste materials (processed or raw) to produce energy. WTE plants include incinerators that produce steam for district heating or industrial use, or that generate electricity; they also include facilities that convert landfill gas to electricity.

Kinds of Wastes

- ❑ **Solid wastes:** domestic, commercial and industrial wastes especially common as co-disposal of wastes

Examples: *plastics, containers, bottles, cans, papers, scrap iron, and other trash*

- ❑ **Liquid Wastes:** wastes in liquid form

Examples: *domestic washings, chemicals, oils, waste water from ponds, manufacturing*

Classification of Wastes according to their Properties

❑ Bio-degradable

can be degraded (paper, wood, fruits and others)

❑ Non-biodegradable

cannot be degraded (plastics, bottles, old machines, cans, containers and others)

Classification of Wastes according to their Effects on Human Health and the Environment

❑ Hazardous wastes

waste that is reactive, toxic, corrosive, or otherwise dangerous to living things and/or the environment. Many industrial by-products are hazardous.

❑ Non-hazardous

Substances safe to use commercially, industrially, agriculturally, or economically.

From where these comes???

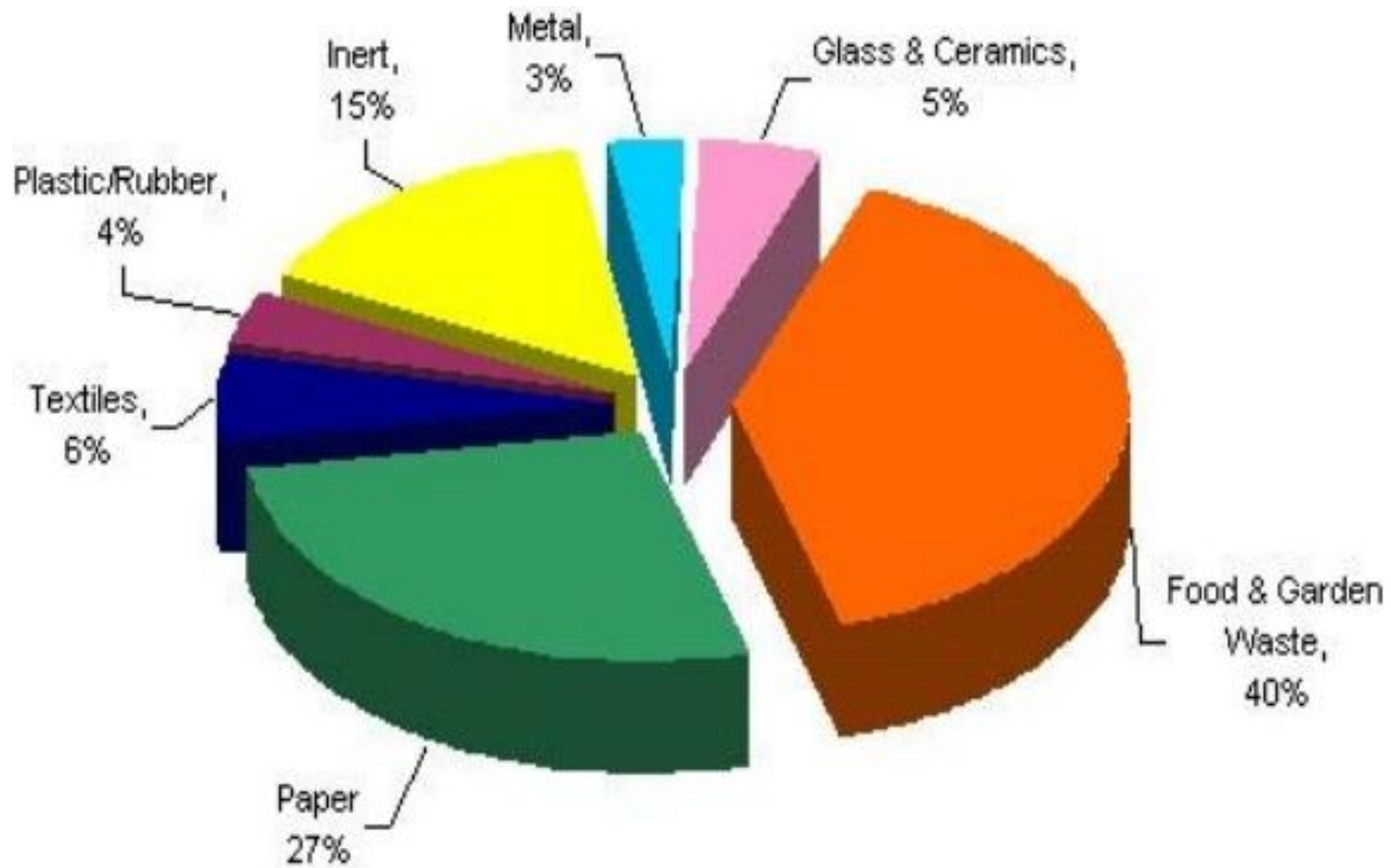


Effects of waste if not managed wisely

- ❑ Affects our health
- ❑ Affects our socio-economic conditions
- ❑ Affects our coastal and marine environment
- ❑ Affects our climate







SOLID WASTE

MANAGEMENT

- **Solid Waste Management** is defined as the discipline associated with control of generation, storage, collection, transport or transfer, processing and disposal of solid waste materials.
- Solid waste management includes planning, administrative, financial, engineering and legal functions in the process of solving problems arising from waste materials.
- The primary goal of solid waste management is reducing and eliminating adverse impacts of waste materials on human health and environment to support economic development and superior quality of life.

Solid waste collection and transport

Factors considered:

i) Types of Containers:

- Depend on:
 - characteristics of SW collected
 - E.g. Large storage containers (Domestic SW: flats/apartment)
 - Containers at curbs
 - Large containers on a roller (Commercial/Industrial)
- Collection frequency
- Space available for the placement of containers

ii) Container Locations:

- side/rear of house
- alleys
- special enclosures (apartment/condos)
- Basement (apts. in foreign countries)/
newer complexes

iii) Public Health:

- relates to on-time collection to avoid the spread of diseases by vectors, etc.

iv) Aesthetics:

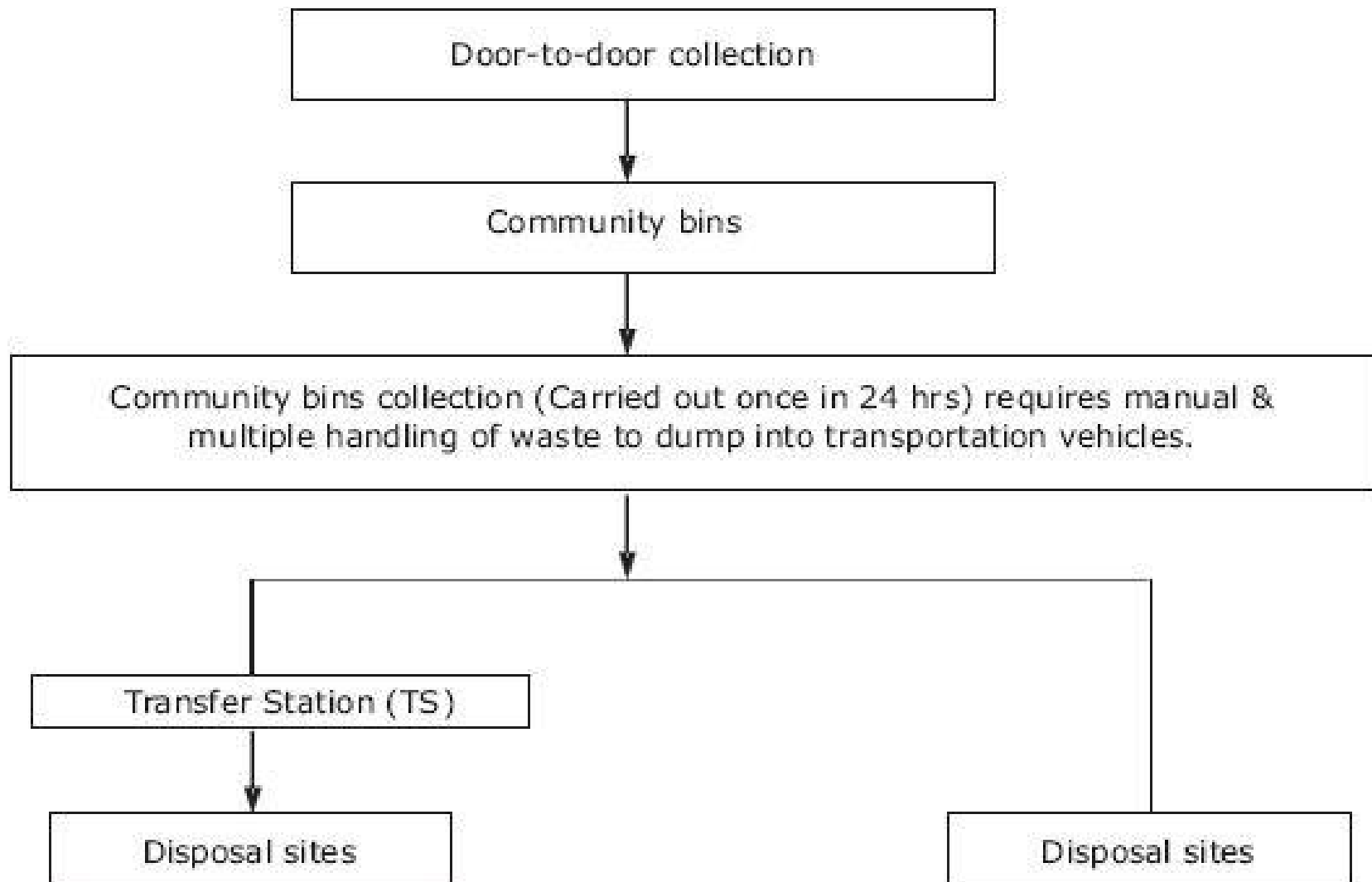
- must be pleasing to the eye (containers must be clean, shielded from public's view).

v) Collection of SW

- 60-80 percent of total SWM costs.
- Malaysia (other developing nations) - labor and capital intensive.
- Major problems:
 - Poor building layouts - e.g. squatters
 - Road congestion - time cost, leachate, transport costs.
 - Physical infrastructure
 - Old containers used (leaky/ damaged)
 - Absence of systematic methods (especially at apartments, markets with large wst. volume).

Collections were made by:

1. Municipal/ District Council
2. Private firm under contract to municipal
3. Private firm contract with private residents



Types of collection

Municipal Collection Services:

- a. Residential:
 - 1. Curb (*Kerb-side*)
 - 2. Alley
 - 3. Set out and set back
 - 4. Backyard collection

Curb (Kerb-side)

- ❑ House owner is responsible for placing solid waste containers at the curb on scheduled day.
- ❑ The work man come, collect and empty the container and put back at the curb.
- ❑ House owner is required to take back the empty containers from the curb to his house.
- ❑ Quickest/ economical
- ❑ Crew: 1 driver + 1 or 2 collectors
- ❑ No need to enter property

Set-out, set back

- ❑ Collectors have to enter property
- ❑ Set out crew carries full containers from resident storage location to curb/ alley before collection vehicle arrives.
- ❑ Collection crew load their refuse into vehicle
- ❑ Set-back crew return the container to storage area.

Alley service

- The containers are placed at the alley line from where they are picked up by workmen from refuse vehicles who deposit back the empty container.

Backyard service

- ❑ The workers with the vehicles carry a bin, wheel
– barrow or sack or cloth to the yard and empty the solid waste container in it.

- ❑ The bin is taken to solid waste vehicles where it is emptied.

Commercial-Industrial Collection Services:

- i. Large movable and stationary containers
- ii. Large stationary compactors (to form bales)

Collection Frequency:

- ❑ residential areas :everyday/ once in 2 days
- ❑ communal/ commercial : daily
- ❑ food waste - max. period should not exceed :

- the normal time for the accumulation of waste to fill a container
- the time for fresh garbage to putrefy and emit fould odor
- the length of fly-breeding cycle (< 7 days).







BLUE BOX



YELLOW BAG



BLUE BAG

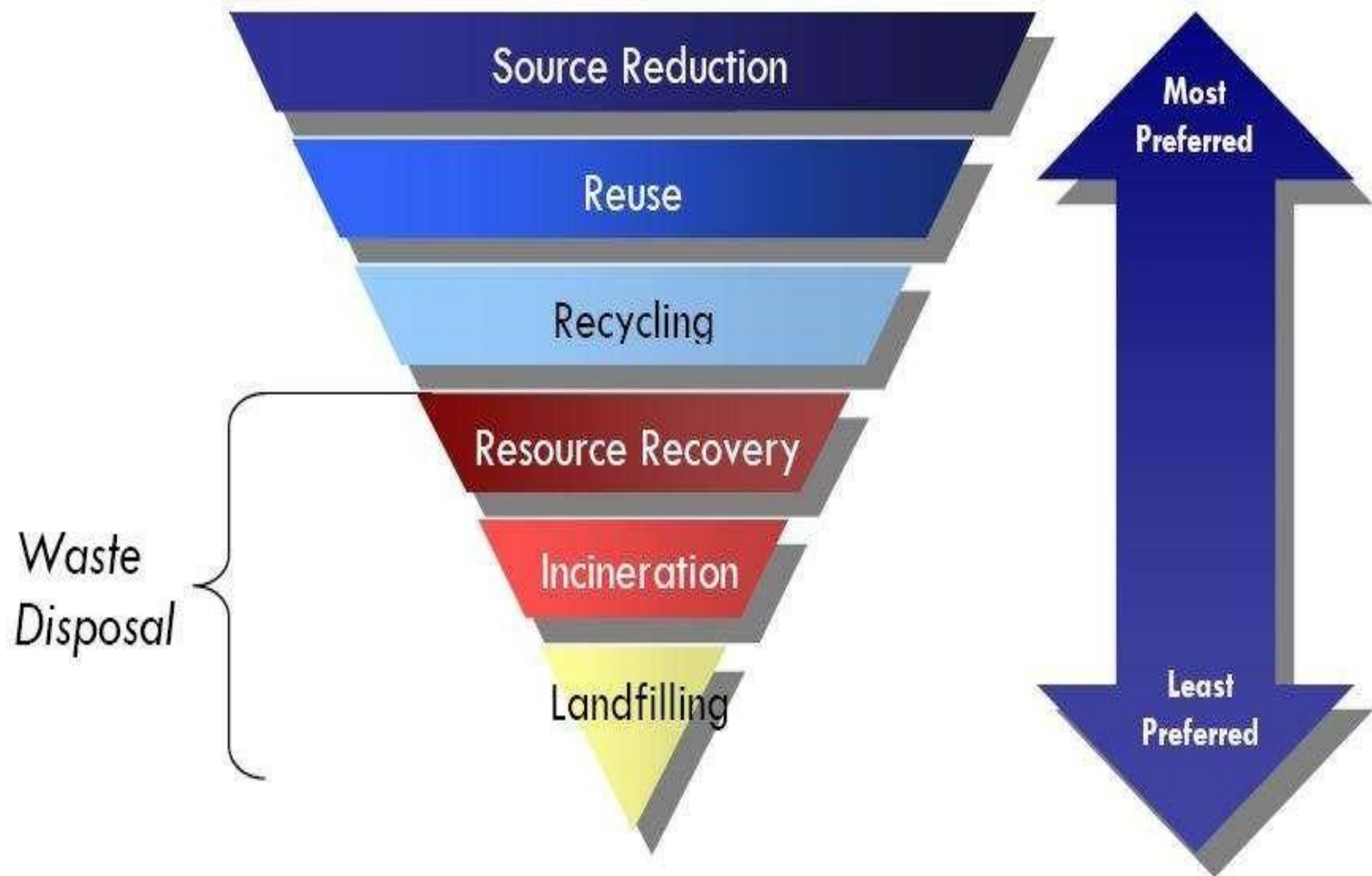
All material accepted in the Recycling Program is banned from the landfill.

Treatment and disposal of solid waste

Several methods are used for treatment and disposal. These are:

1. Composting
2. Incineration
3. Landfilling
4. Pyrolysis
5. Recycling

Figure 3-1. The Solid Waste Management Hierarchy



Composting

- ❑ It is a process in which organic matter of solid waste is decomposed and converted to humus and mineral compounds.
- ❑ Compost is the end product of composting, which used as fertilizer.
- ❑ Three methods of composting:
 - (a) composting by trenching
 - (b) open windrow composting
 - (c) mechanical composting

Composting by trenching

- ❑ Trenches 3 - 12 m long, 2 – 3 m wide and 1- 2 m deep with spacing 2 m.
- ❑ Dry wastes are filled up in 15 cm. On top of each layer 5 cm thick sandwiching layer of animal dung is sprayed in semi liquid form.
- ❑ Biological action starts in 2- 3 days and decomposition starts.
- ❑ Solid waste stabilize in 4- 6 months and changed into brown colored odorless powdery form known as humus.



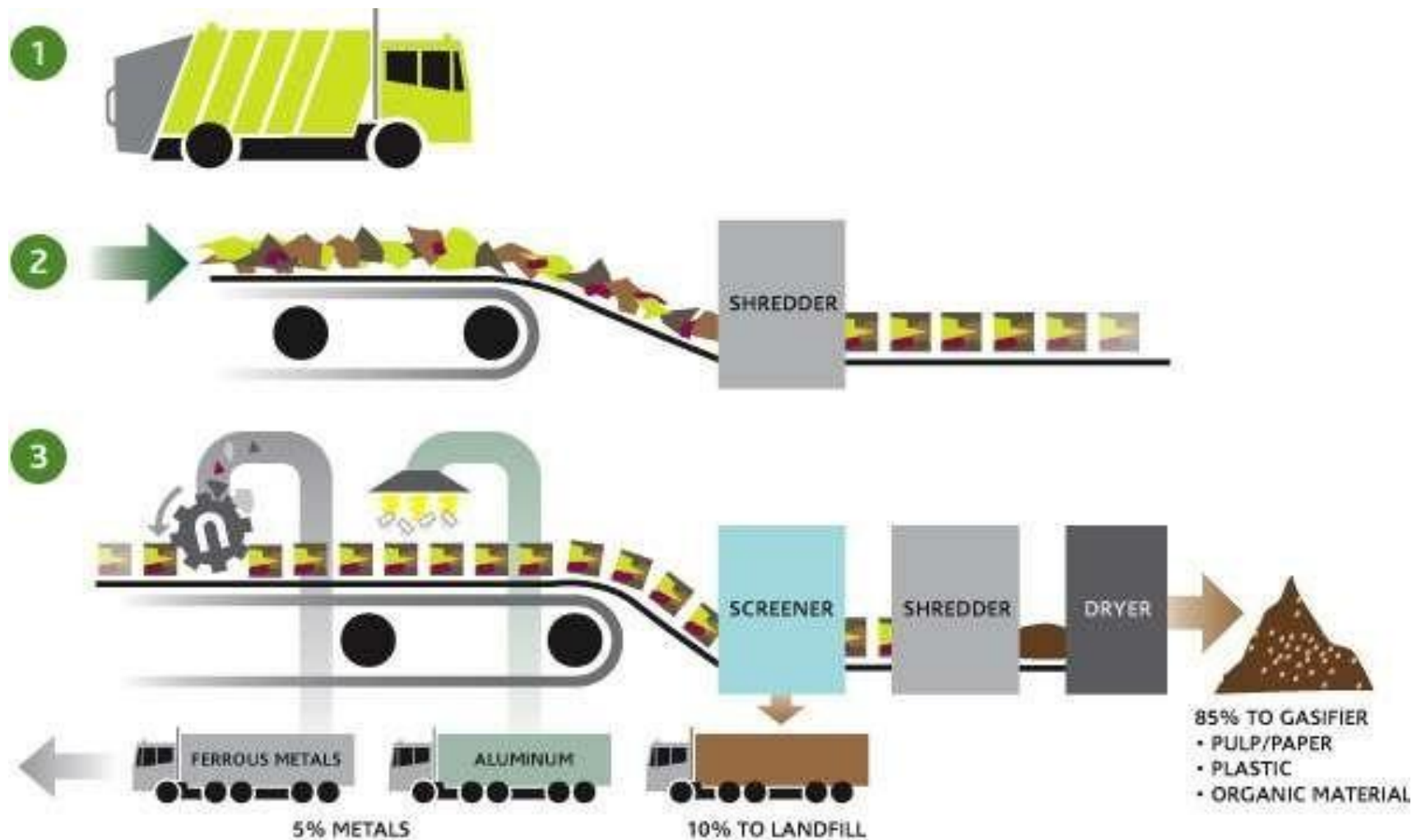
Open window composting

- ❑ Large materials like broken glass, stone, plastic articles are removed.
- ❑ Remaining solid wastes is dumped on ground in form of piles of 0.6 – 1 m height.
- ❑ The width and length of piles are kept 1- 2 m and 6 m respectively.
- ❑ Moisture content maintained at 60%.
- ❑ Temp. increases in side pile.
- ❑ After pile for turned for cooling and aeration to avoid anaerobic decomposition.
- ❑ The complete process may take 4- 6 week.



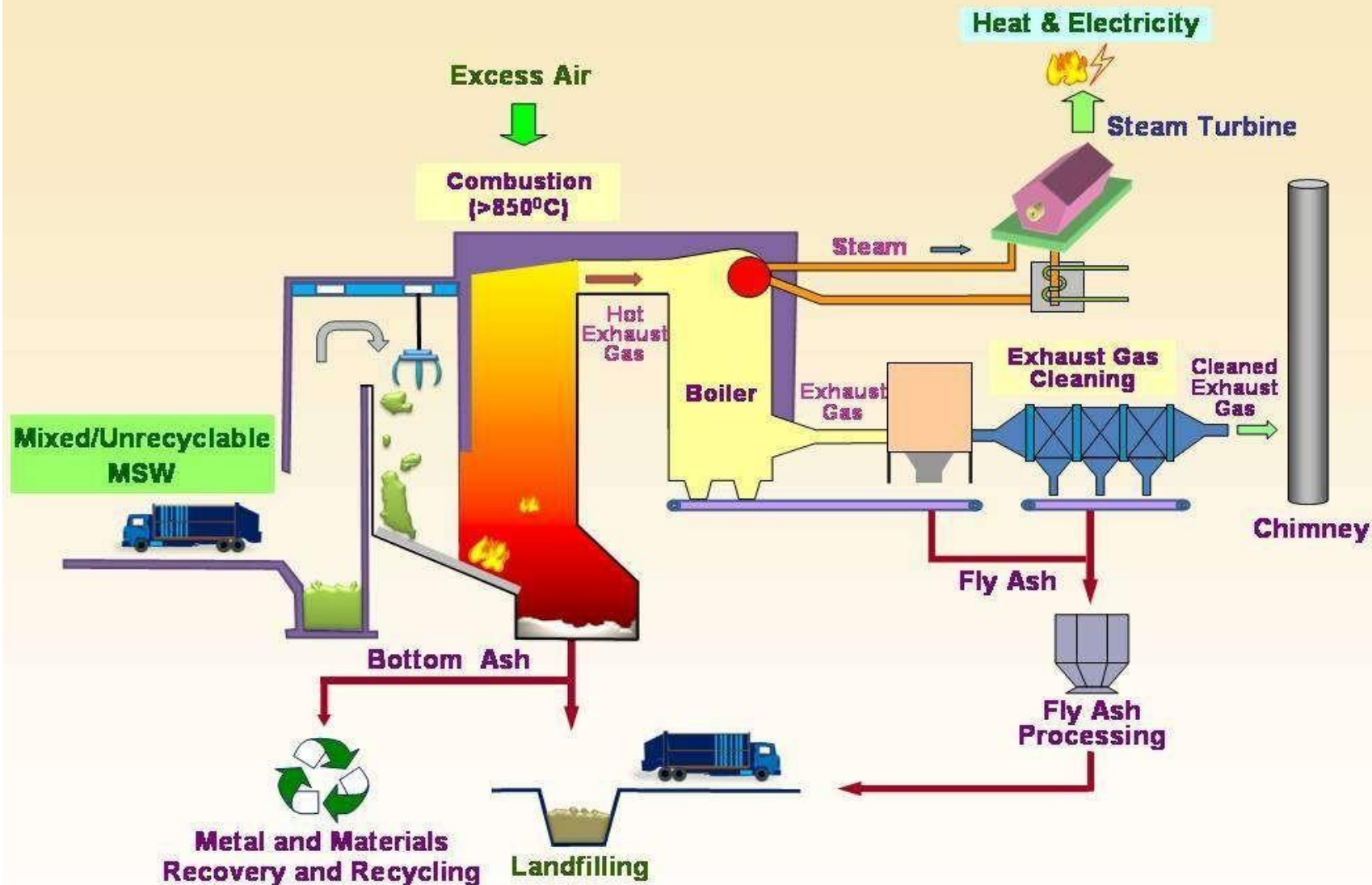
Mechanical composting

- ❑ It requires small area compare to trenching and open windrow composting.
- ❑ The stabilization of waste takes 3- 6 days.
- ❑ The operation involved are
 - reception of refuse
 - segregation
 - shredding
 - stabilization
 - marketing the
 - humus



Incineration

- ❑ Incineration is a waste treatment process that involves the combustion of organic substances contained in waste materials.
- ❑ Incineration and other high temperature waste treatment systems are described as "thermal treatment".
- ❑ Incineration of waste materials converts the waste into ash, flue gas, and heat.
- ❑ Incinerators are used for this process.



Important points regarding incineration

- ❑ Supplying of solid waste should be continuous.
- ❑ Waste should be proper mixed with fuel for complete combustion.
- ❑ Temp. should not less than 670 °C.

Advantages

- ❑ Most hygienic method.
- ❑ Complete destruction of pathogens.
- ❑ No odor trouble.
- ❑ Heat generated may be used for steam power.
- ❑ Clinkers produced may be used for road construction.
- ❑ Less space required.
- ❑ Adverse weather condition has no effect.

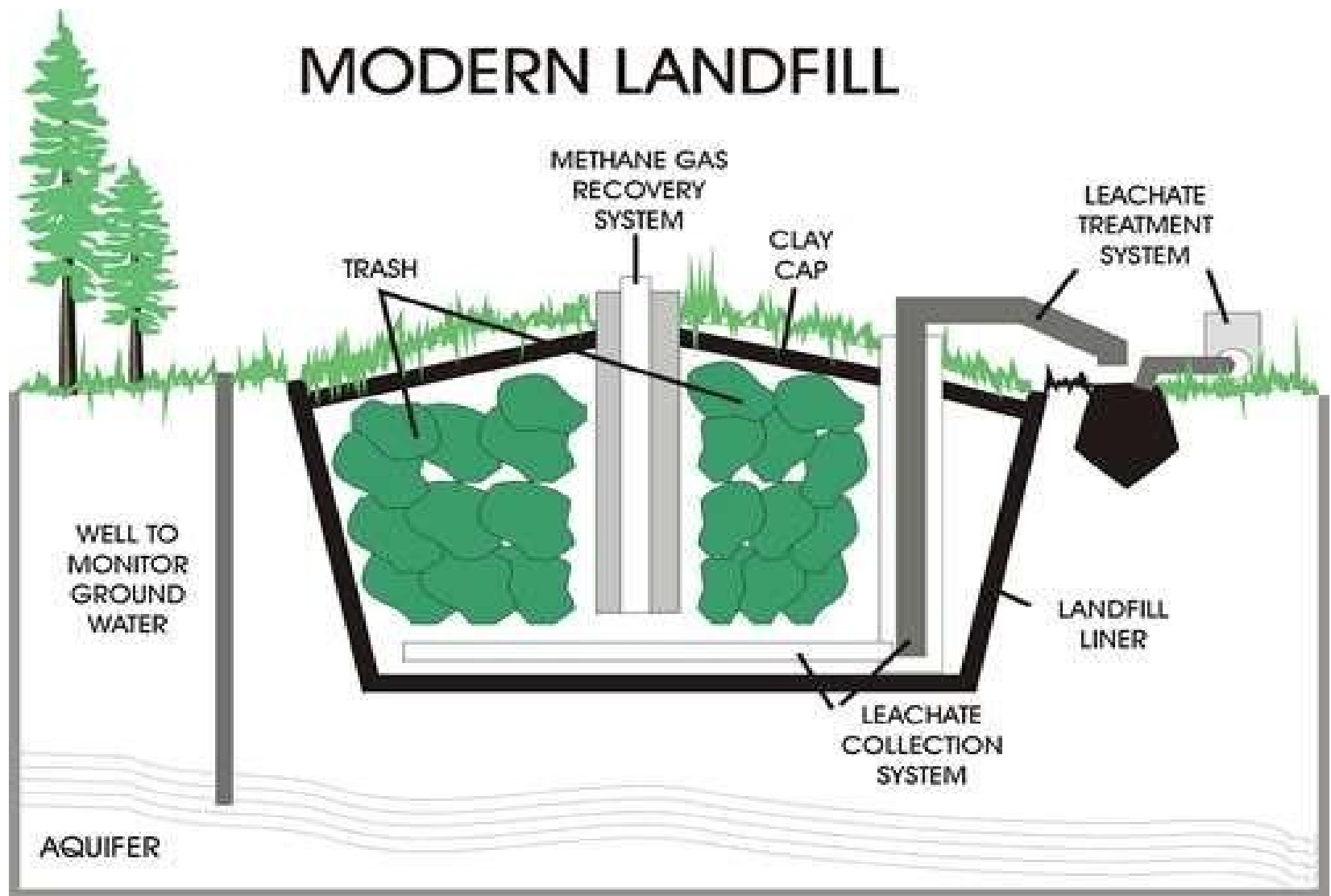
Disadvantages

- ❑ Large initial expense.
- ❑ Care and attention required otherwise incomplete combustion will increase air pollution.
- ❑ Residues required to be disposed which require money.
- ❑ Large no of vehicles required for transportation.

Landfilling

- ❑ A landfill site is a site for the disposal of waste materials by burial and is the oldest form of waste treatment.
- ❑ Historically, landfills have been the most common methods of organized waste disposal and remain so in many places around the world.
- ❑ The dumping is done with layers of 1- 2 m.
- ❑ The layer is covered with soil of 20 cm thickness.

MODERN LANDFILL



Advantages

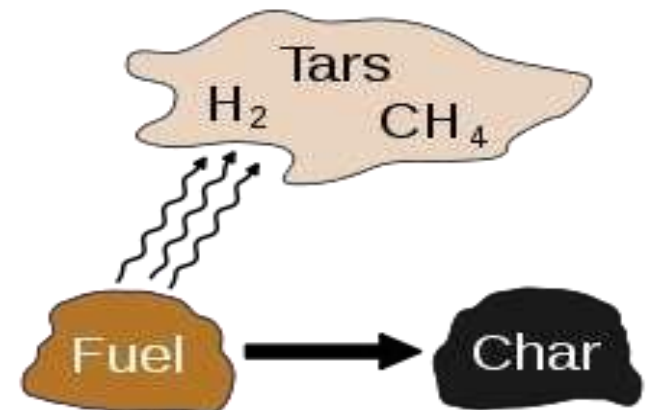
- ❑ Simple method.
- ❑ No costly plant required.
- ❑ No residues or by products need to be disposed.
- ❑ Separation not required.
- ❑ Unused land can be used.
- ❑ Methane gas can be used as fuel.

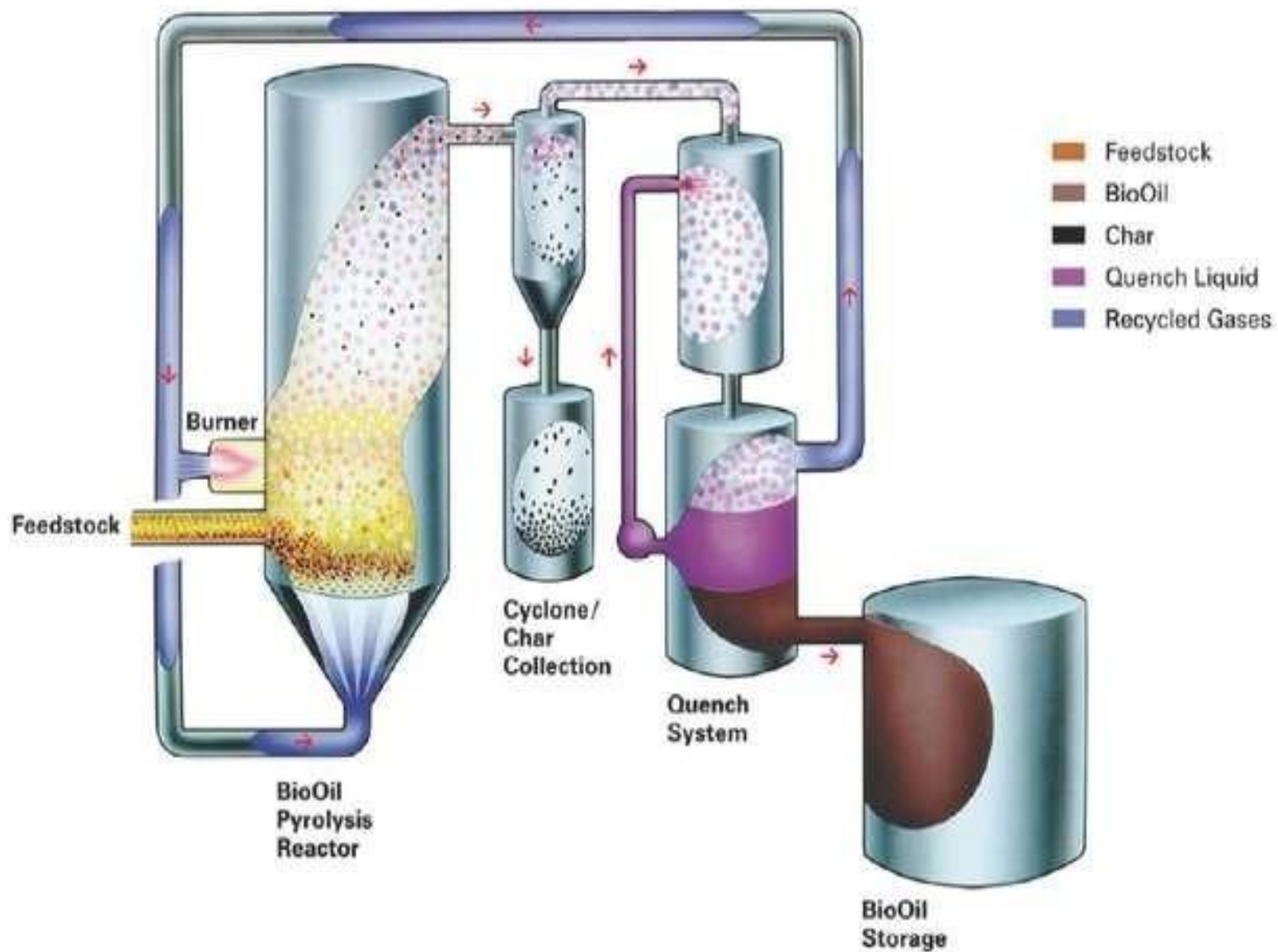
Disadvantages

- ❑ Large land required.
- ❑ Proper dumping site may not be available.
- ❑ Odor problem.
- ❑ Use of insecticides required.
- ❑ Leachate should be collected regularly.
- ❑ Methane gas should be collected properly.
- ❑ Green house gas problem.

Pyrolysis

- ❑ Heating of the solid waste at very high temp. in absence of air.
- ❑ Carried out at temp. between 500 °C – 1000 °C.
- ❑ Gas, liquid and chars are the by products.





Recycling

- ❑ Recycling is processing used materials into new products .
- ❑ It reduce the consumption of fresh raw materials, reduce energy usage, reduce air pollution (from incineration) and water pollution (from landfilling).
- ❑ Recycling is a key component of modern waste reduction and is the third component of the "Reduce, Reuse, Recycle" waste hierarchy.

- ❑ Recyclable materials include many kinds of glass, paper, metal, plastic, textiles, and electronics.
- ❑ Although similar in effect, the composting or other reuse of biodegradable waste – such as food or garden waste – is not typically considered recycling.
- ❑ Materials to be recycled are either brought to a collection centre or picked up from the curbside, then sorted, cleaned, and reprocessed into new materials.

WASTE RECYCLING





Thank u ...